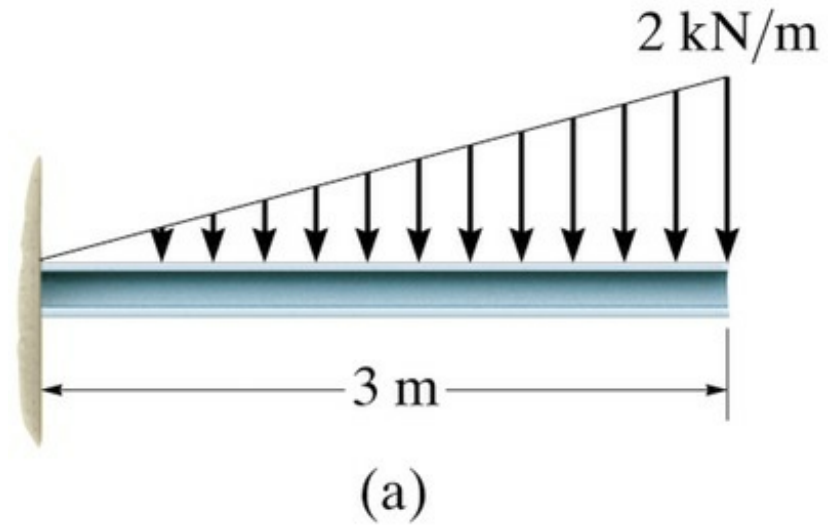


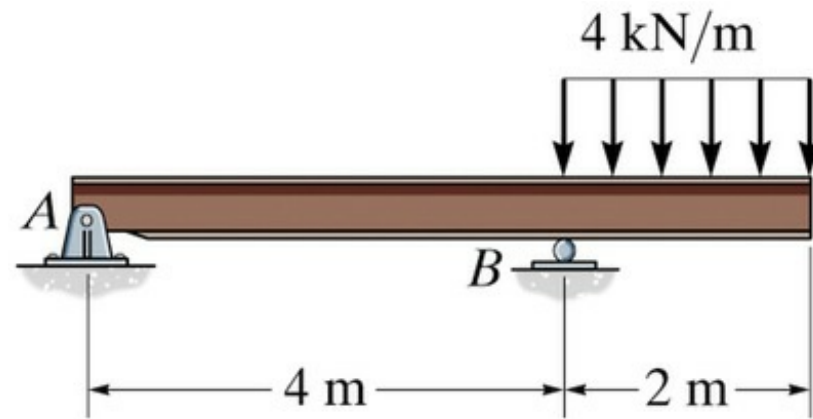
1.



Determine the moment reaction in kN.m at the support for the cantilever beam shown.

2.

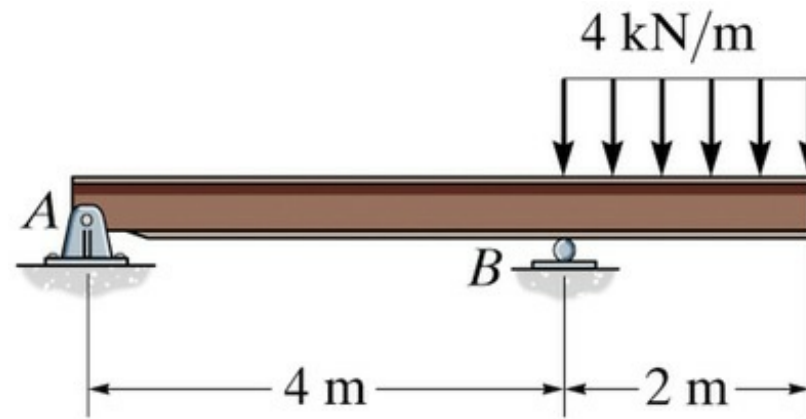
Determine the support reaction at A in kN, include the positive or negative sign to indicate direction of the force.



(a)

3.

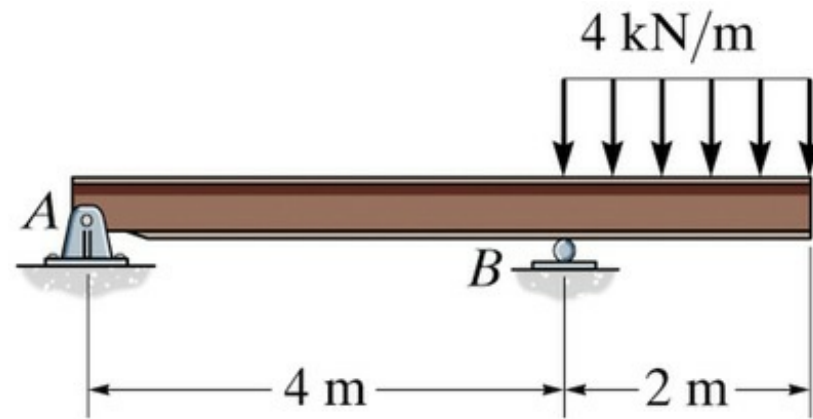
Plot the shear diagram for the beam shown and determine the maximum value for shear in kN, and include signs in your answer.



(a)

4.

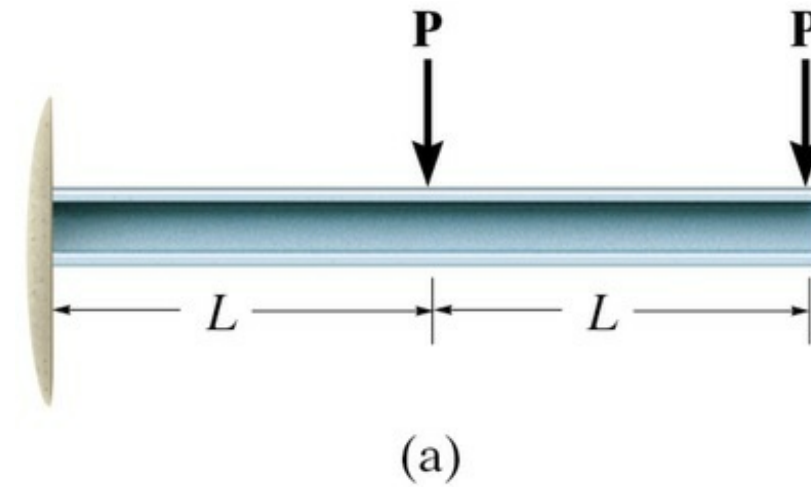
Plot the Shear and Moment diagram for the beam shown and determine the maximum value for the moment in kN.m, include signs in your answer.



(a)

5.

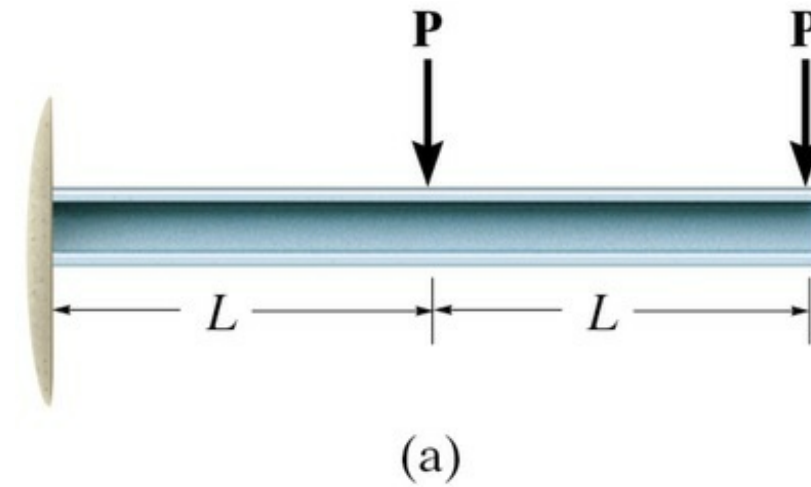
Determine the reaction force at the support.



- A. P
- B. $2P$
- C. $-P$
- D. $-2P$
- E. 0

6.

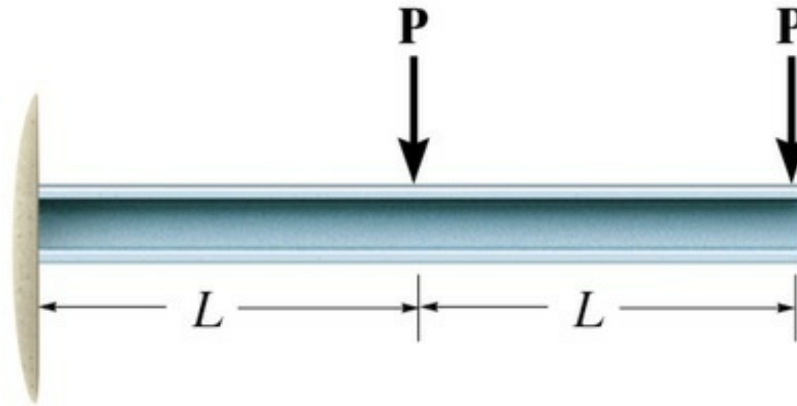
Determine the moment reaction at the support



- A. $+3PL$
- B. $-2PL$
- C. $-PL$
- D. $+2PL$
- E. $+PL$

7.

Draw the shear and moment diagram for the cantilever beam and read the maximum shear force within the beam, make sure to include signs.



(a)

8.

A simply supported beam of length 4 m has a 20 kN point load in the centre. What is the reaction force at each support?

A. 2.5 kN

B. 5.0 kN

C. 10 kN

D. 20 kN

9.

A simply supported beam of length 4 m has a point load in the centre. What is the maximum bending moment in the beam?

- A. 5 kN.m
- B. 10 kN.m
- C. 20 kN.m
- D. 2.5 kN.m

10.

A cantilever beam of 2 m length has a 5 kN downward force at its free end. What is the shear force along the length of the beam?

- A. $V=0$ kN
- B. $V=2.5$ kN
- C. $V= 5$ kN
- D. V varies linearly from 0 to 5 kN

11.

A simply supported beam of length $L=6$ m, carries a uniformly distributed loading (UDL) $\omega = 2$ kN/m. What is the maximum bending moment in the beam?

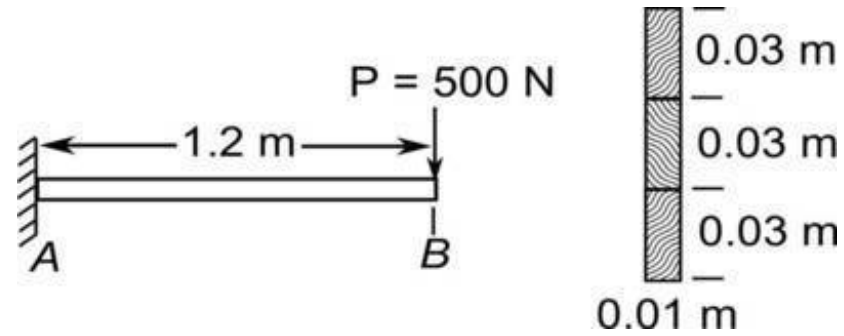
- A. 6 kN.m
- B. 9 kN.m
- C. 12 kN.m
- D. 18 kN.m

12.

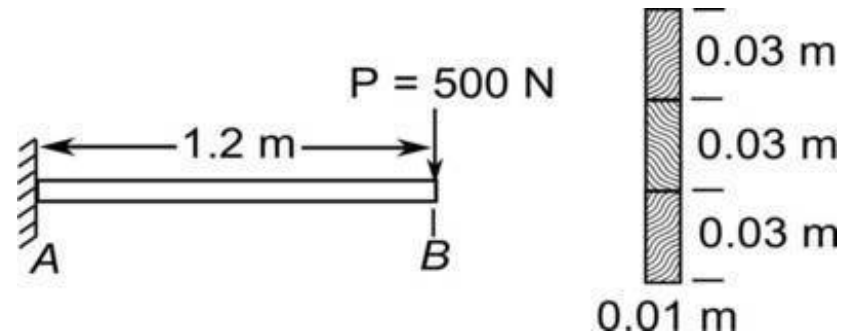
A rectangular beam with cross sectional dimensions, 40 mm wide and 80 mm tall is subjected to a bending moment of 2 kN.m. What is the maximum bending stress in the beam?

- A. 12 kN.m
- B. 24 kN.m
- C. 47 kN.m
- D. 60 kN.m

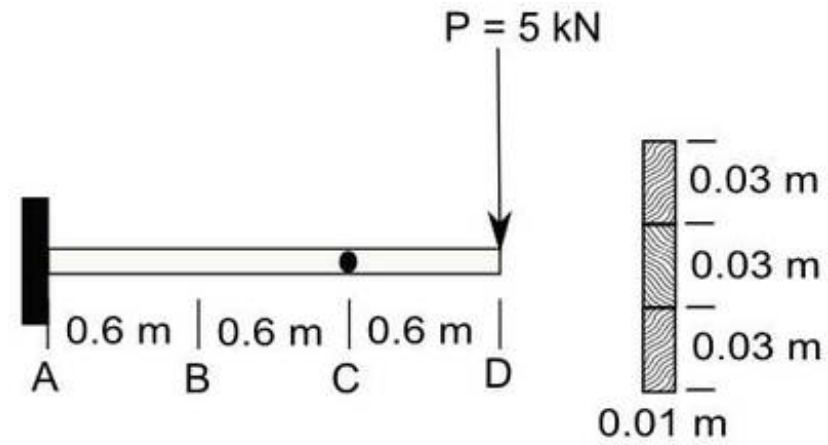
13.



Consider the beam loaded as shown and its cross-section. Using your pointing device place an asterisk (*) at the point on the cross-section where the maximum compressive normal stress due to bending will occur.



14.



In units of MPa, what is the maximum compressive normal stress due to bending at a point 1.2 m from the left end of this beam (location C) ?

15.

An I-beam has the following dimensions:

Flange width: 100mm

Flange thickness: 10 mm

Web height (between flanges) : 80 mm

Web thickness: 10 mm.

The beam is made of steel and is subjected to a bending moment $M = 5 \text{ kN.m}$ acting about the centroidal axis (i.e., causing tension at the bottom and compression at the top).

Determine the moment of inertia of the cross-section about the horizontal centroidal axis J_x in mm^4 .

16.

An I-beam has the following dimensions:

Flange width: 100mm

Flange thickness: 10 mm

Web height (between flanges) : 80 mm

Web thickness: 10 mm.

The beam is made of steel and is subjected to a bending moment $M = 5 \text{ kN.m}$ acting about the centroidal axis (i.e., causing tension at the bottom and compression at the top).

You already determined the moment of inertia of the cross-section about the horizontal centroidal axis I_{xx} in mm^4 . Now determine the maximum bending stress in the beam in MPa.